

```
Processando config 4, seed 0, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config4_seed0.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells =  $2^L$  = 274877906944
Expected number per cell =  $1 / 32768$ 
EColl =  $n^2 / (2k)$  = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density =  $n / k = 1 / 32768.00$ 
Expected number of collisions =  $\mu = 128.00$ 
```

```
-----
Results of CollisionOver test:
```

```
POISSON approximation :
Expected number of collisions =  $N \cdot \mu$  : 256.00
Observed number of collisions : 222
p-value of test : 0.98
```

```
-----
Total number of cells containing j balls
```

```
j = 0 : 549739036894
j = 1 : 16776772
j = 2 : 222
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

```
-----
CPU time used : 00:00:04.53
```

```
Generator state:
16777216 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config4_seed0.bin
```

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 17.00  
p-value of test : 0.39

-----  
CPU time used : 00:00:00.06

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 8.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.07

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : -0.70  
p-value of test : 0.76

-----  
CPU time used : 00:00:00.01

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 6.47  
p-value of test : 0.49

-----  
Normal statistic for number of jumps : 1.65  
p-value of test : 0.05

-----  
CPU time used : 00:00:00.18

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : 0.99  
p-value of test : 0.16

-----  
CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : -1.85  
p-value of test : 0.97

-----

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.019  
p-value of test : 0.69

Kolmogorov-Smirnov- statistic = D- : 0.024  
p-value of test : 0.55

Anderson-Darling statistic = A2 : 0.34  
p-value of test : 0.90

-----

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 7.83  
p-value of test : 0.05

-----  
Global longest run of 1 : 22.00  
p-value of test : 0.63

-----  
CPU time used : 00:00:00.06

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 100.72  
p-value of test : 0.30

-----  
CPU time used : 00:00:00.01

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 18.08  
p-value of test : 0.70  
-----

CPU time used : 00:00:00.01

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -1.39  
p-value of test : 0.92  
-----

CPU time used : 00:00:00.01

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : 0.42  
p-value of test : 0.34  
-----

CPU time used : 00:00:00.01

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----  
Normal statistic : 6.73e-3  
p-value of test : 0.50

-----  
CPU time used : 00:00:00.01

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 175.52  
p-value of test : 0.52

-----  
CPU time used : 00:00:00.01

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 289.18  
p-value of test : 0.73  
-----

CPU time used : 00:00:00.01

Generator state:

235945504 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 477.74  
p-value of test : 0.63  
-----

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :    1.23
p-value of test          :    0.11
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config4_seed0.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -0.41
p-value of test          :    0.66
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config4_seed0.bin
```

```
sstring_Run test:
```

```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355443
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 41.54
p-value of test : 0.24
-----
```

```
-----
Total number of bits: 13417440
```

```
Normal statistic for number of bits : -0.84
p-value of test : 0.80
```

-----  
CPU time used : 00:00:00.04

Generator state:  
299694528 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 2.01  
p-value of test : 0.57

-----  
CPU time used : 00:00:00.06

Generator state:  
316471744 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 1056.73  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.07

Generator state:  
333162944 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 24.02 |
| p-value of test              | : | 0.99  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 47.33 |
| p-value of test              | : | 0.26  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 64    |
| ChiSquare statistic          | : | 50.34 |
| p-value of test              | : | 0.89  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 36    |
| ChiSquare statistic          | : | 43.16 |
| p-value of test              | : | 0.19  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 22    |
| ChiSquare statistic          | : | 23.46 |
| p-value of test              | : | 0.38  |

-----  
CPU time used : 00:00:00.06

Generator state:  
349940160 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 94      |       |
| ChiSquare statistic          | : | 253.24  |       |
| p-value of test              | : | 1.1e-16 | ***** |

-----  
Test on the values of the Statistic M

|                              |   |        |       |
|------------------------------|---|--------|-------|
| Number of degrees of freedom | : | 94     |       |
| ChiSquare statistic          | : | 177.67 |       |
| p-value of test              | : | 4.2e-7 | ***** |

-----  
Test on the values of the Statistic J

|                              |   |        |  |
|------------------------------|---|--------|--|
| Number of degrees of freedom | : | 512    |  |
| ChiSquare statistic          | : | 460.75 |  |
| p-value of test              | : | 0.95   |  |

-----  
Test on the values of the Statistic R

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 90    |  |
| ChiSquare statistic          | : | 75.05 |  |
| p-value of test              | : | 0.87  |  |

-----  
Test on the values of the Statistic C

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 50    |  |
| ChiSquare statistic          | : | 52.57 |  |
| p-value of test              | : | 0.37  |  |

-----  
CPU time used : 00:00:00.04

Generator state:

366717376 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config4\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 148.84 |
| p-value of test              | : | 0.12   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 142.92 |
| p-value of test              | : | 0.22   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 154.58 |
| p-value of test              | : | 0.63   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 120.31 |
| p-value of test              | : | 0.72   |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 89     |
| ChiSquare statistic          | : | 113.80 |
| p-value of test              | : | 0.04   |

-----  
CPU time used : 00:00:00.03

Generator state:

383494176 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:06.18  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test  |                          | p-value |
|-------|--------------------------|---------|
| ----- |                          |         |
| 22    | MatrixRank, 320 x 320    | eps     |
| 25    | RandomWalk1 H (L = 1024) | 1.1e-16 |
| 25    | RandomWalk1 M (L = 1024) | 4.2e-7  |
| ----- |                          |         |

All other tests were passed